renewable energy

HOME NEWS WORLD LIFESTYLE SPORTS

RENEWABLE ENERGY

HYDROPOWER

1 Hydropower is created when moving water turns turbines to create electricity. The source of the moving water can be rivers, waterfalls or the sea. Because flowing water continues to move, this creates an **inexhaustible** amount of energy which can be stored and used when the demand is highest. There are a few drawbacks, however. Water-powered turbines can have a negative environmental impact on **aquatic** wildlife and can endanger boats. Also, creating hydropower dams causes land behind the dams to permanently flood. Finally, it is expensive to set up hydropower systems – the average cost is between $1 million and $4 million.

WIND POWER

2 To create wind power, large turbines are placed on top of hills or **offshore**. The wind turns the blades, which **generate** energy. Wind turbines can be **utilized** on a large scale or on a small scale. Unlike hydropower, this process is relatively cheap and is considered one of the most affordable forms of electricity today. Also, it does not harm the air or land it uses. However, many people consider wind turbines ugly and noisy. Also, they rely on the wind, so if it is not windy, no energy is produced. Finally, like hydropower, wind turbines can be a threat to wildlife such as local birds.

SOLAR ENERGY

3 To use solar energy, solar panels absorb sunlight and, using devices called photovoltaic cells, turn it into electricity. The sun produces energy constantly, which makes solar energy an inexhaustible resource. Another benefit is that it generates no pollution. Solar energy can be adapted to work on a variety of buildings and in a variety of environments. However, a large area of land is needed to produce a large amount of solar power. Scientists have determined that if we wanted to try to power the entire Earth with renewable solar power, we would need to cover a land area about the size of Spain with solar panels. In places with less sunlight, solar power generation has limitations. Also, photovoltaic cells are fragile and can be easily damaged.

[1]**greenhouse gases** (n) gases, such as carbon dioxide, which cause a gradual warming of the Earth's atmosphere

BIOMASS ENERGY

4 Biomass is a biologically produced fuel made from plant and animal material, which is mostly composed of carbon, hydrogen and oxygen. It is the oldest source of renewable energy, used since humans first started burning wood for fire. Today, steam from burning biomass – made up of garbage and other organic waste, rather than wood – turns turbines, generating electricity. Biomass can re-grow over a relatively short period of time compared to the hundreds of millions of years that it takes for fossil fuels to form. It is also an efficient way to generate power, and it is **universal**. Another benefit is that it reduces the need to bury garbage under the ground in a landfill. However, some people are concerned that burning biomass contributes to global warming because it produces greenhouse gases[1]. Also, using biomass to generate energy on a large scale can be expensive.

GEOTHERMAL ENERGY

5 With geothermal energy, heat which is trapped in the ground can be converted into steam to turn turbines. The power it generates can then be used to produce electricity and heat buildings. Geothermal energy uses relatively simple technology. Like several of the other energy sources already mentioned, this process causes no pollution and is inexhaustible. The most active geothermal resources are usually found in areas near volcanoes or where geothermal activity naturally occurs. The largest area of this kind is known as the 'Ring of Fire'. It rims the Pacific Ocean and is bounded by eastern Asia and the western edge of the Americas. Outside of regions like these, geothermal energy is usually unavailable. The **initial** costs of installing a geothermal energy system are very high, but once it is built, the running costs are low.

GEOTHERMAL REGIONS OF THE WORLD



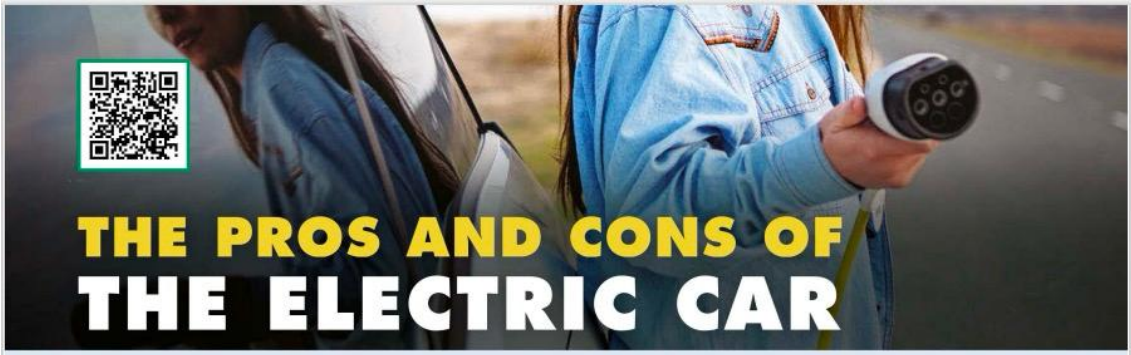
renewable energy	advantage \pros	disadvantage \ cons
水能 hydropower	1. 来源丰富 the source of the moving water can be rivers, waterfalls or the sea --- 持续流动 flowing water continues to move --- 取之不尽 this creates an inexhaustible amount of energy which can be stored and used when the demand is highest	1. 对环境有影响 water-powered turbines can have a negative environmental impact on aquatic wildlife 2. 威胁船只 endanger boats 3. 大坝 creating hydropwer dams --- 洪灾 cause land behind the dams to permanently flood 4. 贵 expensive --- 建议水力发电系统 set up hydropower systems
风能 wind power	1. 便宜 relatively cheap -- 人们可以买得起 be considered as one of the most affordable forms of electricity -- 原因 1： wind turbines can be utilized on a large scale or on a small scale 原因 2: 地理位置 - 偏远 large turbines are placed on top of hills or offshore 2. 不污染土地和空气 does not harm the air or land it uses	1. 噪音污染 wind turbines are noisy 2. 不好看 wind turbines are ugly 3. 不稳定 unstable -- 依靠风 rely om the wind --没有风 it is not windy --没有能源 no energy is produced 4. 对野生动物有威胁 --鸟 wind turbines can be a threat to wildlife -- local birds --涡轮机 turbine --- 桨叶 blade
solar energy 太阳能	1. 取之不尽 solar energy is an inexhaustible resource --- 原因：the sun produces energy constantly 2. 无污染 generate no pollution 3. flexibility 灵活度 - be adapted to a variety of buildings and environment	1. 不稳定 unstable --- 阳光少 in places with less sunlight -- 太阳发电有限 solar power generation has limitations 2. 光伏发电板 photovoltaic cells -- 脆弱 fragile --- 易碎易损 can be easily damaged 3. 耗材很大， 不切实际 impractical --- 支持整个地球 power the entire Earth with renewable solar power --- 太阳能板需要西班牙那么大 --- cover a land area about the size of Spain with solar panels -- 技术不成熟 technology is not so advanced to support
biomass energy 生物质能 definition 定义--	1. 短时间生产 biomass can grow over a relatively short period of time 对比： fossil fuel:	1. 环境影响 --焚烧 burn ---- 温室气体 produce greenhouse gases - carbon dioxide

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burning biomass - garbage, organic waste	petrol\ oil \ coal -the hundreds of millions of years 2. 高效 efficient way to generate power -- 普遍性很强 it is universal 3. 处理垃圾 - reduce the need to bury garbage under the ground in a landfill	emissions--- 全球变暖 burning biomass contributes to global warming 2. 大规模 using biomass on a large scale to generate energy --- 贵 expensive
geothermal energy 地热能	1. 无污染 no pollution 2. 取之不尽 inexhaustible 3. 运营成本低 once it is built, running costs are low 4. 相对简单的科技 relatively simple technology	1. 区域受限 limited regions -- 主要在靠近火山的区域 the most active geothermal resources are usually found in areas near volcanoes or where geothermal activity naturally occurs --- 其他区域地热能不可用 outside of regions like theses, geothermal energy is usually unavailable 2. 初始成本高 initial costs of installing a geothermal energy system are very high
对比		
fossil fuel 化石燃料	来源 source: 耗时很长 fossil fuels are energy sources that are extracted from plants and animals that have been fossilized - or preserved in rock - for hundreds of thousands of years or more -- 天然形成 - 不费钱	化石能源含碳量高 fossil fuels contain high levels of carbon --- 使用的时候二氧化碳和温室气体被排放 carbon dioxide and other greenhouse gases are emitted into the atmosphere -- 空气污染 air pollution

passage 2
the pros and cons of the electric car

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THE PROS AND CONS OF THE ELECTRIC CAR

1 We are on the brink of[1] change for the car industry and countries around the world agree that all cars should be fuelled by an energy source which is more environmentally friendly, such as electricity. Essentially, an electric car is a vehicle powered by a battery, which is charged by an external electricity source. **Unlike** most current cars, which run on[2] fossil fuels, the electric car would not create **extreme** levels of pollution. However, despite this, many countries are delaying this switch to a greener energy source, because whilst this change would bring with it some enormous benefits, it also has some significant **drawbacks**. The following essay will consider both sides of the argument.

2 As already mentioned, perhaps the greatest advantage of electric vehicles (EVs), is that they are kinder to the environment. According to the Mayor of London, road pollution is currently responsible for half the city's air pollution. Unlike gasoline- or diesel[3]-fuelled cars, EVs produce zero exhaust pipe[4] emissions and studies suggest that just one electric car on the road, could save an average of 1.5 million grams of CO2 per year. This idea is further **evidenced** by countries such as Norway, where an increased uptake in the use of EVs has significantly reduced the country's level of carbon emissions.

3 Other advantages of EVs is that they perform well and don't require much **maintenance**. According to many car **manufacturers** EVs now outperform comparable diesel- or gasoline-powered cars. Furthermore, EVs are much quieter than **conventional** cars. Gasoline-fuelled cars are noisy due to the process of **combustion** occurring in each of the engine's cylinders. However, this is not necessary with EVs and therefore by comparison the driving experience can feel much smoother and quieter. Moreover, electric cars are more efficient to maintain. This is because the battery motor will not need oil changes, you will not need to replace engine parts, and generally EVs will not experience the same level of damage over time as a gasoline-fuelled car.

4 However, as with all new technologies, EVs do come with some notable **drawbacks**. One of the major difficulties with **banning** all fossil fuel cars is the cost implication. Although in the long term EVs might be cheaper to run, in the short term EVs are still quite expensive to buy. One of the main reasons for this is that the battery which powers the car is costly. It is the most expensive part of an EV because it is still relatively expensive to produce. However, as battery technologies improve, the price is predicted to decrease in the future.

5 Another significant disadvantage of EVs is that electric cars can travel less distance. The average range for an EV is estimated to be 340 km per charge, whereas some luxury models are able to reach 483 km plus per charge. However, these types of cars are expensive. For example, the Tesla Model X, which provides approximately 542 km of range, has a starting price of about £90,000. By comparison, the average gasoline or diesel vehicle can reach 483km on a full tank, and more fuel-efficient cars can achieve a much further distance. Fuelling an all-electric car can also be problematic. Fully recharging the battery pack can take up to eighty hours, and even fast charging stations can take 30 minutes to charge to 80% capacity. The location and availability of EV charging points is also a concern for those wanting to purchase an electric car. Although the EV charging infrastructure is expanding all the time, in many countries charging points are still not sufficiently accessible. Consequently, owners of EVs must plan their journeys carefully.

6 In conclusion, whilst owning an EV is clearly much better for the environment and will most probably become mandatory in the future, currently the initiative is still experiencing teething problems. However, once these technologies have been developed further, then EVs will have a very positive impact on societies worldwide.

[1]**on the brink of** (idiom) it is likely that a situation will happen soon
[2]**run on** (phr v) to use a particular supply of power
[3]**diesel** (n) a type of fuel
[4]**exhaust pipe** (n) the pipe at the back of the vehicle through which waste gas escapes from the engine

both views 的题目 - 不站立场： the following essay will consider both sides of the argument.

争执点： an electric car \ vehicle

enormous benefits	significant drawbacks
环保 environmentally friendly \ are kind to	价格高 cost implication -- 运行便宜 be

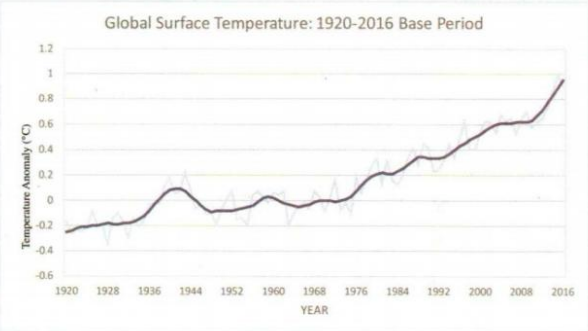
the environment -- 用电 powered by a battery \ an external electricity source --- 不是化石能源 do not run on fossil fuels - 进一步有的解释 柴油汽油车 gasoline - or - diesel - fuelled cars -- 无废弃排放 produce zero exhaust pipe emissions --不会产生很大的污染 will not create extreme levels of pollution 论证逻辑链 air pollution 空气污染 -- 主要来源是道路污染 road pollution is responsible for half the city's air pollution --- 柴油汽油车 gasoline - or - diesel - fuelled cars -- 电车: electric vehicles: 无废弃排放 produce zero exhaust pipe emissions - 降低二氧化碳排放量 reduce the level of carbon emissions	cheaper to run -- 但是: 购买成本高 expensive to buy in the short term --- 电池很贵 battery is costly \ expensive to produce - 更换 exchange
更安静 are much quieter than conventional cars --- 对比论证: gasoline - fuelled cars are noisy -- 噪声来源 - 燃烧: the process of combustion occurring in each of the engine's cylinder 气缸 - 体验感好 good driving experience	行驶的里程数比较短 electric cars can only travel less distance - 豪车可以跑的远 some luxury models are able to reach long distance per charge- 但是贵 but these cars are expensive- 普通车: be estimated to be 340-540 kms per charge -----对比 by comparison, 油车 traditional cars with full tank can achieve a much further distance
好养护 are more efficient to maintain \ do not require much maintenance ---用电池 the battery motor will not need oil changes -- 不需要更换发动机 will not need to replace engine parts --- 电车没有油车损伤那么大 not experience the same level of damage over time as a gasoline - fuelled car	充电时间是个问题 fuelling an all-electric car can be problematic---充满要几小时 fully recharging---take up 80 hours---快充也要 30 分钟且只充满 80% even fast charging take 30 minutes to charge to 80% capacity
	charging point 充电桩 - location \ availability - not sufficient - not convenient - plan carefully

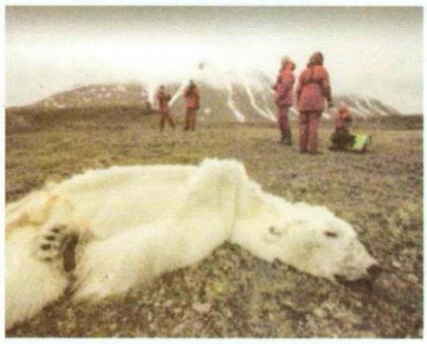
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passage 3: environmental protection (课后去读)

CLIMATE CHANGE REQUIRES THE WORLD'S ATTENTION

We have known about climate change for decades. There is little doubt that Earth is getting warmer and warmer (see the graph). A warming ocean and atmosphere along with melting ice and rising sea levels provide evidence of a dramatic change in the global climate.





In 2013, a lot of people were shocked by a news photo of a dead polar bear that was found on Norway's Arctic island of Svalbard. According to the scientists who found its dead body, all that remained of the polar bear was "skin and bones". An expert who has studied polar bears for many years said that from the position of its dead body, the bear appeared to have **starved** and died. Experts claimed that low sea-ice levels caused by climate change meant the bear could not hunt seals as before, so it had to travel greater distances in order to find food. This alarming case showed how the increase in temperature had an impact on Earth's **ecology**.

Then what is causing the increase in the global average surface temperature? Climate scientists often mention a key climate process called the "greenhouse effect", which has two common meanings: the "natural" greenhouse effect and the "man-made" greenhouse effect. The "natural"

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greenhouse effect refers to the fact that heat from the sun enters the atmosphere and warms Earth's surface as short-wave radiation. The heat is **released** back into space at longer wave lengths. Greenhouse gases in the atmosphere, such as methane and **carbon** dioxide, trap some of the heat, keeping Earth's climate warm and habitable. Without this process, Earth could not **sustain** life. However, the "man-made" greenhouse effect has now become a big problem. When people produce huge amounts of extra greenhouse gases by burning fossil **fuels**, more heat energy is trapped in the atmosphere and causes Earth's surface temperature to rise quickly.

There is strong and **comprehensive** evidence that the rise in temperature has led to an increase in extreme weather and natural disasters worldwide, not only causing serious damage, but also costing human lives. Climate scientists have warned that if we do not take appropriate actions, this warming **trend** will probably continue and there will be a higher price to pay. In fact, news reports are **frequently broadcast** about extreme rainstorms and heatwaves causing deaths and economic losses.

Continued greenhouse gas emissions will result in further warming and long-lasting changes to the global climate. This requires the attention of people all over the world. Governments need to consider making **policies** and taking appropriate actions and measures to reduce greenhouse gas emissions. We as individuals can also reduce our "carbon footprint" by **restricting** the amount of carbon dioxide our lifestyles produce. It is our responsibility to **seize** every opportunity to educate everyone about global warming, along with its causes and impacts, because this is the most serious issue affecting all of us on this planet. So what will you do to help?

passage 4:

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AIR POLLUTION AND PUBLIC HEALTH

1 It may be a familiar sight: a dull haze lingers over a city. The haze has a brownish quality and limits visibility, even as people go about their daily lives. While this may be a common scenario in some cities, it is important to know that air pollution in every form, whether visible as smog (a combination of the words smoke and fog) or invisible, poses a global threat to human health. Smog may not be present in every city, but data from the World Health Organization indicates that 99% of the world's population breathes polluted air. Globally, air pollution is one of the leading causes of chronic disease and premature death; in some countries, it contributes to up to 10% of all deaths. There are several causes of air pollution, though they all stem from human activity.

2 Fossil fuels are energy sources that are extracted from plants and animals that have been fossilized – or preserved in rock – for hundreds of thousands of years or more. These extracts are converted into such energy sources as electricity, heat and gasoline. Everything from cars to homes to small businesses to gigantic industries relies on fossil fuels to generate energy. However, fossil fuels contain high levels of carbon, allowing carbon dioxide and other greenhouse gases to be emitted into the atmosphere. This creates air pollution.

3 How, then, does air pollution impact health? Take one example: fossil fuel powered vehicles. Cars, trucks and buses release poisonous chemicals into the air in the form of particulate matter, which is inhaled into the body by the simple act of breathing. And because the matter is smaller than a grain of sand, it can easily make its way into the bloodstream or lungs. This triggers **adverse** effects, including breathing difficulties and lung infections. People with existing health conditions are especially vulnerable to **complications** that arise from air pollution. Prolonged exposure can lead to lung cancer, heart disease and serious respiratory illnesses. In short, air pollution impacts the global **mortality** rate, a fact that should raise the alarm for immediate action to be taken.

4 Another cause of air pollution is wildfires. Across the globe, climate change is evident. Temperatures are rising, resulting in warmer seasons that last longer and a ground that is drier than ever. One spark, whether from a carelessly thrown cigarette, a smouldering[1] campfire or a strike of lightning, can ignite a wildfire. The resulting smoke carries toxic particulate matter for potentially thousands of kilometres, exposing countless people to air pollution. Wildfire smoke also releases carbon monoxide. Breathing in this poisonous gas prevents sufficient oxygen from reaching the body's organs and tissues. Consequences include dizziness and nausea. People also report feeling fatigue or 'brain fog' during wildfire season. One study has found that particulate matter can impact the attention span after only a few hours of exposure.

5 It is imperative that action be taken to **counter** the effects of air pollution. It is a health hazard for all humankind, but especially **problematic** for vulnerable populations. A California study found that children who were exposed to the smoky air of wildfires suffered from coughing, bronchitis, colds and other respiratory ailments. A 2020 report from the Centre for Research on Energy and Clean Air concluded that, as a result of air pollution, 16 million children live with asthma.

6 Reducing air pollution is key to improving public health. Countries like the United States, where 81% of the total energy is derived from fossil fuels, must **phase out** non-renewable energy sources and instead utilize sustainable sources such as solar and wind power. In the face of recent world events, reducing wildfires is urgent. In 2023 alone, over 800,000 acres of land burned in Chile, Canada experienced its most devastating fires in recorded history, and Spain reported that their wildfire season now lasts longer than ever. The United Nations Environment Programme provides a **grim** warning, predicting that extreme wildfires will have increased by up to 50% by the end of the 21st century.

7 Air pollution is closely intertwined with public health; therefore, reducing the former will improve the latter. For this reason, it is essential to take steps now, especially if we want future generations to not only survive but **thrive**. Children should be able to grow up breathing clean air rather than being at the mercy of a dying planet.

[1]**smouldering** (v) burning slowly; producing smoke but no flames

考核频率	默写检查	中文	英文
		adj. 隐隐的、萧条的	dull
		n. 雾霾	haze
		v. 逗留	linger
		常见的场景	common scenario
		慢性病	chronic disease
		adj. 早产的， 提早的， 仓促的	premature

		a 导致了 b (不是强因果)	a contribute to b
		来源于	stem from
		被改为， 被转化为	be converted into
		n. 汽油	gasoline
		大企业	gigantic industries
		adj. 有毒的	poisonous
		n. 化学物质	Chemicals
		adj. 微粒的 n. 微粒状物质	particulate
		adj. 尤其的	particular
		v. 激发	trigger
		负面影响	adverse effect
		对...很脆弱	be vulnerable to
		现有的身体状况	existing health condition
		并发症	complication
		长时间置身于	prolonged exposure
		呼吸道系统疾病	respiratory illnesses
		简而言之	in short
		死亡率	mortality rate
		v. 闷然	smoulder
		篝火	campfire
		v. 点燃	ignite
		有毒的微粒子	toxic particulate
		一氧化碳	carbon monoxide
		n. 组织	tissue
		眩晕	dizziness
		n. 呕吐	nausea
		n. 疲惫	fatigue
		脑雾	brain fog
		注意力长短	attention span
		汽油	gasoline
		逐步淘汰，分阶段停止使用	phase out
		不同步	out of phase
		紧急的	urgent
		更人担忧的，渺茫的	grim
		ab 紧密交织在一起	a is closely intertwined with b
		v. 繁荣	thrive

空气污染的理由

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化石能源 fossil fuels - 含碳量高 contain high level of carbon --排放二氧化碳和温室气体 emit carbon dioxide and other greenhouse gases into the atmosphere - 空气污染 air pollution

化石能源如何影响身体健康 - 可以拆成两个逻辑
论证污染： xxxxxx --- 污染
论证影响： 污染 ---- 影响
化石能源 ----交通工具 fossil fuel powered vehicles: cars\ trucks \ buses ----释放有毒的化学物质 release poisonous chemicals into the air in the form of particulate matter ---通过呼吸被吸入人体 be inhaled into the body by the simple act of breathing --- 进一步论证：粒子比沙子小 the matter is smaller than a grain of sand -- 很容易进入血液和肺部 easily make its way into the bloodstream or lungs--- 对身体有负面影响 trigger adverse effect on human health --- 尤其是脆弱人群 especially problematic for vulnerable populations: children and senior citizens - 举例论证： 呼吸困难 breathing difficulties \ 肺部感染：lung infections \ 并发症 complications that arise from air pollution \ 肺癌 lung cancer \ 心脏病 heart disease \ 呼吸道系统疾病 respiratory illness \ailment \ 咳嗽 coughing \ 哮喘 asthma\ 支气管炎 bronchitis ---- 死亡率 mortality rate

air pollution 的原因是野火 wildfire -- 全球气温上升 temperatures across the globe are rising ---- 干燥的季节持续很长 warmer seasons last longer --- 地面更加干燥 ground is drier than ever --- 小火花可能会导致野火 one spark can ignite a wildfire -- 举例论证：随手丢的烟头 carelessly thrown cigarette \ 闷火 smouldering campfire \ 一道闪电的袭击 a strike of lightning ---- 烟雾 smoke --- 携带有毒的微粒 carry toxic particulate matter --- 烟雾释放一氧化碳 release carbon monoxide --- 空气污染 expose countless people to air pollution -- - 对身体有负面影响 it is a health hazard for all humans--- 具体的展开：呼吸了有毒的气体 breathing in the poisonous gas -- 阻止了氧气进入到身体的器官和组织 prevent sufficient oxygen from reaching the body's organs and tissues --- 眩晕和恶心 dizziness and nausea \ 疲惫 fatigue \ 脑雾 brain fog \ 注意力 short attention span

解决措施 it is imperative that action be taken to counter the effects of air pollution \ it is essential to take steps to counter the effects of air pollution --- 1.放弃使用传统能源 phase out non- renewable energy sources --- 使用新能源 utilize sustainable sources -- 举例： solar energy \ wind power 2. 减少火灾 reduce wildfires is urgent -- 缩短火灾季 shorten wildfire season

误导且通过禁用你信息牟思大作文环保相关

passage 5

HOW OCEAN POLLUTION AFFECTS HEALTH

1 In many ways, the world's oceans pose a great mystery. Covering more than 70% of the earth, they remain largely unexplored and unseen by human eyes. Nevertheless, oceans around the globe are polluted by human activity. And while it may be easy to blame it on the fishing or shipping industries or on beaches piled with rubbish by too many tourists, the truth is that 80% of marine pollution originates on land. It is brought about by air pollution from urban factories, disease-carrying bacteria from agricultural regions and the plastic we use on a day-to-day basis. These pollutants not only harm marine life, but also promote the **transmission** of disease to humans. Ocean pollution is **facilitating** a public health crisis.

2 Plastic waste is the leading cause of sea pollution. Because it is lightweight and durable, plastic is used in countless ways in almost every aspect of our lives. To illustrate its pervasiveness, the non-profit Plastic Oceans International estimates that, around the world, one million plastic bags are used every sixty seconds. Many are then thrown away. Research shows that, as of 2019, 460 million tonnes of plastic are produced per year and an astounding 0.5% of that finds its way into the ocean. The problem is so severe that Nikola Simpson of the United Nation's Development Programme warns that, if we don't reduce our **reliance** on plastic, plastic fragments may outnumber the fish in the sea by the middle of the twenty-first century.

3 How does plastic find its way to the ocean? Research suggests it is primarily carried by rivers. Imagine, for example, a plastic water bottle is inadequately disposed of in Budapest. That the landlocked city is 500 kilometres away from the coast makes no difference; its **proximity** to a river does. The river carries it out to sea. Along the way, the plastic decomposes, absorbing pesticides and other pollutants that were already present in the water. The plastic eventually breaks down into microplastics, categorized as fragments that are less than 5 millimetres in length, small enough for fish and shellfish to consume. Humans then eat the contaminated fish, allowing our bodies to also take in those tiny, toxic fragments. Research on the consequences of ingesting microplastics, which has been called 'the silent menace', points to cause for concern. A 2023 study in *Environment & Health* found evidence that microplastics can damage the liver, intestines and brain; they may also damage DNA. Additional research published in 2023 for the California State Legislature listed endocrine, digestive and reproductive problems, as well as respiratory illnesses and cancer, as some of the health hazards of microplastics.

4 Another dangerous ocean pollutant is mercury, a naturally occurring element that comes from coal. When coal is burned as an energy source, mercury settles into the atmosphere. With the help of rain or snow, it is washed into rivers and carried to the ocean. Over time, fish may accumulate large amounts of mercury in their systems, which is then delivered to humans via the contaminated fish. As a neurotoxin, mercury can damage the nervous system. It may also harm the digestive and immune systems, as well as the lungs, kidneys, skin and eyes. Mercury poisoning puts adults at risk of **incurable** diseases such as heart disease and dementia. Tragically, the danger is not **confined to** adults. If an expectant mother consumes mercury, it can harm the developing brain of her unborn baby. The World Health Organization includes mercury in their top ten chemicals of public health concern.

5 Work must be done to reduce ocean pollution; however, the trend is heading in the wrong direction. Based on a 2023 study in the journal *PLOS One*, there has been a **surge** in ocean plastic pollution since 2005; that research estimated that approximately 171 trillion plastic particles are now present in the world's oceans. The United Nations Environment Programme is pushing for an 80% reduction in plastic use over the next 20 years. On the part of consumers, reusing and recycling are major steps toward reaching that goal. However, real change must occur at the root. Our global reliance on plastic must be minimized, forcing a decrease in production. Once that happens, we can begin the work of **eradicating** the illnesses that arise as a result.

考核频率	英文	中文	默写栏
	shipping industry	货运业	
	bacteria	细菌	
	region	区域	
	facilitate	v. 促进	
	lightweight	adj. 轻的	
	pervasive	adj. 遍布的	

	pervasiveness	n. 无处不再的	
	astounding	adj. 震惊的	
	severe	adj. 严重的	
	fragment	n. 碎片	
	outnumber	v. 超过...的数量	
	inadequate	adj. 不足够的	
	landlocked	adj. 内陆的	
	proximity	n. 附近	
	decompose	v. 分解	
	absorb	v. 吸收	
	microplastic	n. 微塑料	
	take in	吸收	
	menace	n. 威胁	
	liver	n. 肝脏	
	intestine	n. 肠道	
	endocrine	n.内分泌	
	digestive		
	reproductive	adj. 繁殖的	
	respiratory	adj. 呼吸的	
	health hazard	健康风险	
	hazard	n. 危险	
	hazardous	adj. 危险的	
	mercury	n. 水银	
	coal	n. 煤	
	contaminate	v. 污染	
	neuron	n. 神经元	
	toxic	n. 毒素	
	nervous system	神经系统	
	immune system	免疫系统	
	kidney	肾脏	
	incurable	adj. 难以治愈的	
	dementia	老年痴呆	
	be confined to	被限制在	
	trillion	万亿	
	particle	n. 粒子	
	reliance	n. 依赖	
	eradicate	v. 根除	
	at the root	从根源上	

读写互通资源帮你搞定雅思作文环保和能源

海洋污染的原因是人类活动 oceans around the globe are polluted by human activity ---- 展开： 举例论证：渔业 fishing \ 货运业 shipping industries \ 游客乱扔垃圾 beaches are piled with rubbish by too many tourists --- 进一步的论证： 并列 - 其他原因： 来源于陆

地 marine pollution originates on land -- 具体的阐述： 空气污染 - air pollution -- 来自于工厂 urban factories \ 农业活动带来的细菌 disease - carrying bacteria from agricultural regions\ 白色污染 the plastic we use on a day - to - day basis ---- 伤害了海洋生物 harm marine life --- 对人类健康的影响 health crisis: 促进了疾病的传播 promote transmission of disease to humans

塑料污染 plastic waste 影响了海洋 is the leading cause of sea pollution -- 具体展开 ：孰料袋的特质：轻 lightweight 、耐用 durable --- 人们一致在使用 plastic is used in countless ways in almost every aspect of our lives ----塑料无处不在 pervasiveness --- 一河流带到了海里 it is primarily carried by rivers----部分进入海洋 find its way into\to the ocean -- 后果：塑料碎片会超过鱼的数量 plastic fragments may outnumber the fish in the sea

读了第三段话完善本段话的逻辑链条-- 延展到了对人的健康

塑料污染 plastic waste 影响了海洋 is the leading cause of sea pollution -- 具体展开 ：孰料袋的特质：轻 lightweight 、耐用 durable --- 人们一致在使用 plastic is used in countless ways in almost every aspect of our lives ----塑料无处不在 pervasiveness --- 一河流带到了海里 it is primarily carried by rivers----1. 部分进入海洋 find its way into\to the ocean -- 后果：塑料碎片会超过鱼的数量 plastic fragments may outnumber the fish in the sea 2. 塑料会分解 plastic decomposes - 吸收杀虫剂 absorb pesticides and other pollutants that were already present in the water ---- 分解为微塑料 break down into microplastics ----- 碎片很小 small fragments ------ 鱼类会吞食他们 fish and shellfish consume -- 人会吃鱼 humans eat the contaminated fish --- 人体就会吸收这些有毒的碎片 our bodies take in those tiny and toxic fragments -- 对身体形成影响 health hazard\ menace --- 举例： 脑子 brain，肝脏 liver，肠道 intestine，损伤 DNA-- damage DNA，内分泌 endocrine，消化道 digestive，繁殖 reproductive，呼吸道 respiratory illness，癌症 cancer

水银 mercury 是海洋污染物之一 : another dangerous ocean pollutants is mercury ---来源： 煤炭 coal -- 燃烧 burned as an energy source ---- 水银就留存在了空气中 mercury settles into the atmosphere --- 下雨下雪 with the help of rain or snow -- 冲刷到了河流 it is washed into rivers ---- 携带到了海洋 is carried to the ocean --- 鱼吃了水银 fish may accumulate large amounts of mercury in their systems -- 人吃了 delivered to humans via the contaminated fish --- 神经系统 damage nervous system \ 消化系统 harm the digestive system\ 免疫系统 compromise the immune system \ 肺部 lung \ 肾脏 kidney \ 皮肤 skin \ 眼睛 eyes \ 心脏病 heart disease \ 老年痴呆 dementia --- 对成年人 adults \ 婴儿 infants --- 大脑 harm the developing brain of a baby

解决措施 work must be done to reduce ocean pollution --- 减少塑料的生产和使用 reduction in plastic use and production -- 3R: reuse, recycle, reduce -- 减少对于塑料制品的依赖 minimize the reliance on plastic

passage 6

这马上通海燕帮你搞定雅思大作文环保和能源

Disaster mitigation

1 The world has always had to face water-based natural disasters such as tsunamis and hurricanes. In an illuminating[1] interview, *Water Management Monthly* talks to Dan Smith, who works in disaster mitigation for a government agency.

2 'Dan, could you tell us what disaster mitigation means?'

'Disaster mitigation means attempting to minimize the impact of natural disasters both before and after they happen. My department and I work in two specific areas to try and do this: risk **reduction** and risk analysis. Both are **crucial** in disaster mitigation.'

3 'What do you mean by risk reduction?'

'Risk reduction means many things. It is not just referring to big engineering projects like dams. Often, small **community** projects can be the most effective means of risk reduction. The main way floods can be prevented is by the construction and **maintenance** of earth-wall defences, called levees. These block the progress of rising water. However, even the best levees can't protect against the **devastating** power of a tsunami. In this case, early-warning systems are lifesavers. They can let people know as early as possible if there is likely to be flooding.'

4 'What types of risk analysis do you do?'

'Firstly, risk analysis concerns flood mapping, where we **identify** the parts of the country which are at most risk from flooding. Secondly, there is mitigation planning, which helps local communities plan for when disaster strikes. Thirdly, we make sure that the country's dams all work properly and are safe. Although many people **criticize** dams because of their environmental impact, they also have many benefits such as hydroelectricity, irrigation, water storage, water sports and, of course, flood control. In terms of a cost-benefit analysis, we are definitely ahead.'

5 'Do you think countries are better prepared now for natural disasters than they were in the past?'

'Definitely. We are constantly developing new flood-prevention solutions. Some examples of these kinds of **measures** include the construction of sea walls and bulkheads[2], which protect the coasts, and the redesign of power stations and subway tunnels in the New York City area after the devastating damage caused by Hurricane Sandy in 2012. In the UK, another good example is the Thames Barrier, a huge engineering project designed to prevent London from flooding.'

6 'Aren't programmes like that very expensive? What lower-cost alternatives are there?'

'Flood prevention does not have to be expensive. Sandbags, for example, can be a highly effective way of stopping flood water.'

7 'Is there any more that can be done, or are we as prepared as we can be?'

'There's always more that could be done. But remember that the government can only be responsible for flood prevention up to a certain point. People have to become aware of the dangers of flooding themselves. This is crucial. Expensive early-warning systems are a waste of money if people pay no attention to them.'

[1] **illuminating** (adj) giving you new information about a subject

[2] **bulkhead** (n) an underwater wall

passage 7

GOING GREEN IN DENMARK

1 In today's world being environmentally friendly should be high on the agenda[1] for all countries. Every time a natural disaster such as flooding occurs, we are reminded of just what a fragile state our environment is in. Whilst many nations are indeed making great efforts to prevent the negative impacts of climate change there is a small group of countries which are really leading the way in terms of 'green' initiatives. Countries such as: Switzerland, Japan, Thailand and Denmark. In this article, we will consider Denmark's remarkable commitment to protecting the environment, as in 2022 it was ranked first in the Environmental Performance Index (EPI.

2 Historically, Denmark is a country based on agriculture and fishing, and therefore its connection to nature has always been strong. Perhaps this is the reason why the Danes have become **pioneers** at promoting sustainability and have made protecting the environment an integral part of their culture. The country has a long history of green **initiatives.** Through decades of sustained effort, the Danes have invested in: green transportation, renewable energy, water management and waste recycling.

3 One key way in which Denmark strives to protect the environment is by investing in and promoting cleaner modes of transport. Vehicles contribute significantly to CO_2 **emissions** and therefore this is a key area in which countries need to invest. Danes have therefore fostered a cycling culture and in Copenhagen it is said there are more bikes than cars. Denmark has provided its citizens with a complex network of green cycle routes, thus encouraging people to use their bikes instead of their cars. Vehicles are still prominent in Denmark, but cars which are of low or zero emission, are common. This is supported by the fact that there is a fuelling **infrastructure** in place for electric, hydrogen and natural gas vehicles.

4 Another key feature of Danish culture is their commitment to the use of renewable energy sources. By using cleaner forms of energy Denmark aims to reduce carbon emissions, and decrease the emission of **greenhouse gases**. In fact, Denmark aims to be free of fossil fuels by 2050 and rely solely on renewable energy sources such as wind, solar and geothermal[2] power. Wind power has a particularly prominent place in Danish culture, and this began after the oil crisis in 1973, when investment in wind turbines became a necessity. Since then, the use of these turbines has become common, and the government has plans to construct one of the world's first energy islands. This initiative would utilize the abundant wind energy resources in the North and Baltic Seas. It is projected that these offshore wind turbines will have the capacity to power at least 5 million households.

5 Water management is another key global environmental challenge the world faces. **Droughts** are becoming more prevalent, and more people than ever are struggling to get access to clean drinking water and sanitation[3]. However, this is certainly not the case in Denmark, which provides some of the cleanest tap water in the world. The way in which this is achieved is also environmentally friendly, as drinking water comes entirely from groundwater, which is carefully protected and managed **sustainably**.

6 The final area to be considered is Denmark's management of waste. Danes take the amount of waste they produce very seriously. For example, the levels of recycling in Denmark are very high and consumers can use vending machines to return their bottles and cans in exchange for money. Reducing the amount of food waste is also another area the Danes have heavily invested in. Food waste supermarkets, food waste cooking schools and food waste **awareness campaigns** are just some of the initiatives which have helped to foster a culture which prioritizes a reduction in food waste.

7 Whilst no country can be regarded as the winner in the fight against climate change, it is important to recognize those nations who are using **innovation** to rethink the way we do things, in order to protect and preserve our environment for future generations. Denmark is without doubt one of those pioneers and it would be neglectful of other countries not to follow suit with some of its green initiatives.

[1]**agenda** (n) important subjects that have to be dealt with
[2]**geothermal** (adj) of or connected with the heat inside the earth
[3]**sanitation** (n) a system of protecting people's health by removing dirt and waste

passage 8 our changing planet

1 Since 1900 the global temperature has gone up by around 1.0 °C. This may not sound like much, but such a small increase is causing sea levels to rise and **threatening** the habitat of many species of plants and animals. An increase of 2 °C in global temperatures could result in extinction for 30% of the world's land species.

2 The Northwest Passage is a sea route which runs along the northern coast of Canada between the Atlantic and Pacific Oceans. In the past, it was often difficult to use because the water was frozen; however, increasing temperatures and the subsequent deglaciation[1] have made it easier for ships to travel through this route. The trouble is that the melting of the ice is leading to loss of habitat for the polar bears and other species which live in this area.

3 Experts predict that global sea levels could rise by 30.5–122 cm by the end of the century. Consequently, some areas that were land a few hundred years ago are now under water, and many low-lying islands may be under water in the future.

4 As a result of the changing **climate**, the world's **ecosystems** are also changing faster than ever before. More than one-third of the world's mangrove forests[2] and around 50% of the world's coral reefs[3] have been destroyed in the last few decades. Forests are being cut down to provide land for food because human population is growing at such a rapid rate. 38% of the land on Earth is now used for growing food. As a result of the higher temperatures and higher levels of carbon dioxide in the **atmosphere**, plants are producing more pollen, which could lead to more cases of asthma, a medical condition which makes it hard to breathe.

5 What is causing climate change? The main **cause** is the huge amount of **greenhouse gases**, such as methane and carbon dioxide (CO_2), in the atmosphere, but the reason for this is the world's population – you and me. As the population increases, more land is needed to provide food and energy. Burning **fossil fuels** for heating, lighting, transport, electricity or manufacturing produces CO_2. Furthermore, humans breathe out CO_2 while trees 'breathe in' CO_2 and produce oxygen, so by cutting down trees, we are increasing the amount of CO_2 in the atmosphere and reducing the amount of oxygen. As a result of human activities, CO_2 levels are now at their highest in 800,000 years.

rate of about 200 m, so the area is covered in water. This is evidence of **global warming**.

6 The biggest challenge we all face is to prevent further environmental disasters. We must do something before it is too late. We need to reduce the amount of CO_2 in the atmosphere. We need to stop burning fossil fuels and start using renewable energy. We can get enough energy from renewable fuels, such as solar energy, hydroelectric energy or wind power, to be able to stop using fossil fuels completely.

[1]**deglaciation** (n) the melting of a glacier

[2]**mangrove forest** (n) large areas of trees and other plants which grow next to oceans, e.g. in Florida and Bangladesh

[3]**coral reefs** (n) colourful underwater ecosystems built by tiny animals

passage 9

读写互通海燕带你搞定

THE CAUSES AND EFFECTS OF DEFORESTATION

1 Forests, which cover almost one-third of the surface of the Earth, produce oxygen and provide homes to plants, animals and humans. These days, many of the world's great forests are threatened by *deforestation* – the process of removing trees from large areas of land. The **destruction** of forests occurs for several reasons; trees are used as fuel or for **construction**, and cleared land is used as pasture[1] for animals or fields for planting food. The main harmful **effects** of deforestation are climate change and damage to animal habitats.

2 The main causes of deforestation are commercial **farming** by big business and farming by local people. Huge commercial farms have taken over large areas of forest in many countries. In Indonesia, for example, industrial **logging** is carried out to clear huge areas for the production of palm oil, while in Brazil, large areas of the Amazon **rainforest** are cleared to grow soy and produce vegetable oil. Local farmers also cut down and burn trees but they clear areas just big enough to graze cattle or grow crops. However, after two or three years, the land can no longer be used, so the farmer moves to another piece of land. Normally, it takes around ten years for cleared land to recover, but in populated areas the land is never allowed to recover. This constant reuse of land leads to heavy erosion[2] – the loss of the top layer of soil which protects the ground. Erosion, in turn, can cause flooding in heavy rain.

3 One serious effect of deforestation is climate change. Normally tropical rainforests help control the Earth's temperature by **absorbing** carbon dioxide. However, recently, the Amazon area, where much of the world's rainforests are found, has been releasing more carbon dioxide than it absorbs. In addition, in areas where deforestation has taken place, the carbon dioxide goes into the atmosphere and traps heat in a process called the *greenhouse effect*. The result is global warming. Increasing global temperatures result in less rain. This causes the rainforests to dry out and leads to fires – which cause more emissions of carbon dioxide. In this way, the rainforests actually contribute to global warming instead of helping to solve it.

4 Forest destruction also has an effect on biodiversity[3]. Deforestation causes the loss of habitats and damage to land where plants and animal species live, leading to the extinction of many species. A decrease in biodiversity threatens entire ecosystems and destroys future sources of food and medicine.

5 In conclusion, damage to the world's forests is leading to changes in the natural environment and causing global warming. Looking to the future, governments should act to plant more trees which will absorb carbon dioxide and protect forests from illegal logging. Otherwise, deforestation on such a large scale is sure to have terrible effects on the environment.

读写互通海燕帝

作业

环保类 - report 类 话题

The natural resources such as oil, forests and fresh water are being consumed at an alarming rate. What problems does it cause? How can we solve these problems?

Many people continue to use cars and motorcycles even though they know that they are bad for environment. Why is this? What can be done to reduce the use of these vehicles?

More and more plastic waste is polluting the world's city, countryside and ocean. What problems will it cause? What measures should be taken to solve these problems?

Some people think government should focus on reducing environmental pollution to help people prevent illnesses and diseases. To what extent do you agree or disagree?

能源类话题

Fossil fuels (coal, oil, natural gas) are the main sources in many countries, but in some countries the use of alternative sources of energy (wind energy and solar energy) are encouraged. To what extent do you think it is a positive or negative development?

The international community must act immediately to ensure all nations reduce consumption of fossil fuels (gas and oil). To what extent do you agree or disagree?

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